॥ वसुधैव कुटुम्बकम् ॥

SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act 1956)

Re-accredited by NAAC with "A" Grade

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Semester: 3 **Year: 2026-27**

Subject Title: Operating Systems Lab	
EXPERIMENT No: 8	
TITLE: case … esac Statements	**DoP**: 31-8-2026

Aim: Implement shell script to demonstrate **case … esac** statements.

Learning Outcome: 1. To understand the **case … esac** statements
2. To demonstrate shell script to implement **case … esac** statements using Linux Commands

Hardware/Software:

This experiment is performed using Ubuntu OS in a virtualized environment within Windows 11. The software used is:

- VMware Workstation 17 Player (Free Virtualization Tool)
- Ubuntu 22.04.1 LTS ISO file
- VMware Tools installed on the virtual machine

Problem Statement:

a) Shell script using case statement to print whether the character entered by the user is vowel or consonant

b) Shell script to print whether the number entered by the user is a single digit number or 2-digit number or 3-digit number

c) Shell script to check the type of file entered by the user

Write a shell script that asks the user to enter a file name and then determines whether it is a regular file, directory, or other type of file using the case statement.

d) Shell script to check whether an entered character is uppercase, lowercase, digit, or special symbol

Write a script that takes a single character as input from the user and prints whether it is:

Uppercase letter (A-Z), Lowercase letter (a-z), Digit (0-9) Special symbol

e) Shell script to determine the type of vehicle based on user input

Write a script where the user enters a vehicle name (e.g., car, bike, truck, bus), and the script outputs the category:
Car/Bike → Personal Vehicle
Truck → Heavy Vehicle
Bus → Public Transport
Any other input → Unknown Category

f) Shell script to identify a fruit based on user input

Write a shell script where the user enters a fruit name, and the script categorizes it as:
Apple, Mango, Banana → Common Fruits
Strawberry, Blueberry → Berries
Coconut, Walnut → Dry Fruits
Unknown Fruit → Category Not Found

g) Shell script to determine the season based on the month entered by the user

Write a shell script where the user enters a month name or number (1-12) and the script prints the corresponding season:
December – February → Winter
March – May → Spring
June – August → Summer
September – November → Autumn

Theory:

The shell acts as an interface between the user and the OS services. It accepts human-readable commands from the user and converts them into a language that the kernel can understand. The shell can be accessed using the terminal.

Shell accepts the command as input from the user and executes it. A shell can also take inputs as a file and execute them in case of executing a bunch of commands routinely. This helps to avoid the repetitive work of typing each command in the terminal. This file containing a bunch of commands is known as a shell script and is saved with the extension of .sh.

The '**case … esac'** statements are used to implement multi-way branching in shell scripts. The "**case**" statement checks a variable against a series of patterns and executes a block of code associated with the first pattern that matches. It is similar to a '**switch case**' statement as used in C and C++.

The general syntax of **case … esac** statement in Shell Scripting includes:

```
case variable in
  pattern1)
    # commands to execute if variable matches pattern1
    ;;
  pattern2)
    # commands to execute if variable matches pattern2
    ;;
  pattern3)
    # commands to execute if variable matches pattern3
    ;;
  *)
    # default commands to execute if variable doesn't match any pattern
```

;;
esac

The '**case**' statement is closed with '**esac**' (which is simply case spelt backwards). Also, each pattern is followed by a list of commands that are executed if the pattern matches. The **;;** at the end of each command list is used to separate the commands from the next pattern. The ***** pattern is a catch-all (similar to the default case) that matches anything that didn't match any of the previous patterns.

Note:
By default, the .sh file may not have executable permission. Hence, using the command chmod +x filename, the file is given executable permission.

#!/bin/bash signifies that the script should be executed using the bash shell

Algorithm:
For performing **case … esac** statements using shell scripting:

1. Define the variable you want to test against patterns.
2. Start the case statement with the keyword case followed by the variable.
3. For each pattern you want to match against, write the pattern in parentheses, followed by a block of commands to execute if the pattern matches, and end the block with ;;.
4. Add a final pattern * to match any input that didn't match the previous patterns. Include commands to execute if this pattern is matched, and end the block with ;;.
5. End the case statement with the keyword esac.

Program:

Steps to Execute the Program:

1. Create a new file and make it executable using the chmod command
2. Edit the file using Text Editor and add the commands required for performing the desired operations using **case … esac** statements.
3. Save and close the file. The file extension should be .sh
4. Run the script in the terminal using ./filename.sh
5. Based on the operations, provide the necessary inputs and the output is generated based on the evaluation of specified pattern in the **case … esac** statements.

Screenshots:

1.

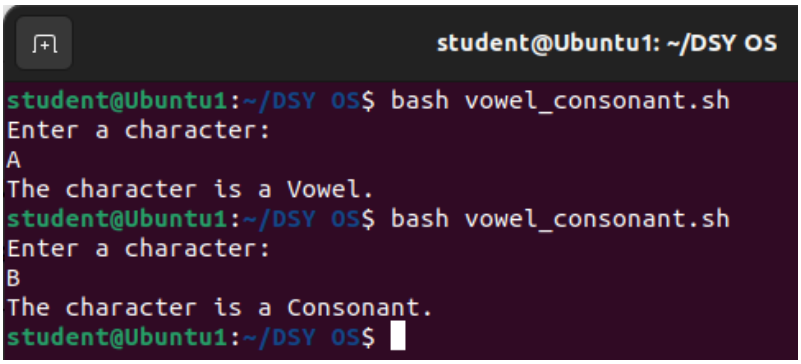
vowel_consonant.sh | number_digits.sh | file_type.sh

```
#!/bin/bash

echo "Enter a character:"
read ch

case $ch in
    [aeiouAEIOU])
        echo "The character is a Vowel."
        ;;
    [a-zA-Z])
        echo "The character is a Consonant."
        ;;
    *)
        echo "Invalid input. Please enter an alphabet."
        ;;
esac

```



2.

vowel_consonant.sh | number_digits.sh | file_type.sh

```
#!/bin/bash

echo "Enter a number:"
read num

case $num in
    [0-9])
        echo "It is a Single Digit Number."
        ;;
    [0-9][0-9])
        echo "It is a 2-Digit Number."
        ;;
    [0-9][0-9][0-9])
        echo "It is a 3-Digit Number."
        ;;
    *)
        echo "It is not a 1, 2, or 3-Digit Number."
        ;;
esac

```

```
student@Ubuntu1:~/DSY OS$ bash number_digits.sh
Enter a number:
17
It is a 2-Digit Number.
student@Ubuntu1:~/DSY OS$ bash number_digits.sh
Enter a number:
123
It is a 3-Digit Number.
student@Ubuntu1:~/DSY OS$
```

3.

file_type.sh × character_type.sh

```
#!/bin/bash

echo "Enter a file name:"
read file

if [ -e "$file" ]; then
    type=$(stat -c %F "$file")

    case $type in
        "regular file")
            echo "$file is a Regular File."
            ;;
        "directory")
            echo "$file is a Directory."
            ;;
        *)
            echo "$file is another type of file."
            ;;
    esac
else
    echo "File or directory does not exist."
fi

```

```
student@Ubuntu1:~/DSY OS$ bash file_type.sh
Enter a file name:
abc.txt
abc.txt is a Regular File.
student@Ubuntu1:~/DSY OS$ bash file_type.sh
Enter a file name:
file.txt
File or directory does not exist.
student@Ubuntu1:~/DSY OS$
```

4.

character_type.sh × vehicle_type.sh

```
#!/bin/bash

echo "Enter a single character:"
read ch

case $ch in
    [A-Z])
        echo "The character is an Uppercase Letter."
        ;;
    [a-z])
        echo "The character is a Lowercase Letter."
        ;;
    [0-9])
        echo "The character is a Digit."
        ;;
    *)
        echo "The character is a Special Symbol."
        ;;
esac

```

```
student@Ubuntu1:~/DSY OS$ bash character_type.sh
Enter a single character:
g
The character is a Lowercase Letter.
student@Ubuntu1:~/DSY OS$ bash character_type.sh
Enter a single character:
L
The character is an Uppercase Letter.
student@Ubuntu1:~/DSY OS$
```

5.

vehicle_type.sh ×

```
#!/bin/bash

echo "Enter vehicle name:"
read vehicle

case $vehicle in
    car|Car|CAR|bike|Bike|BIKE)
        echo "Personal Vehicle"
        ;;
    truck|Truck|TRUCK)
        echo "Heavy Vehicle"
        ;;
    bus|Bus|BUS)
        echo "Public Transport"
        ;;
    *)
        echo "Unknown Category"
        ;;
esac

```

```
student@Ubuntu1:~/DSY OS$ bash vehicle_type.sh
Enter vehicle name:
car
Personal Vehicle
student@Ubuntu1:~/DSY OS$ bash vehicle_type.sh
Enter vehicle name:
truck
Heavy Vehicle
student@Ubuntu1:~/DSY OS$
```

6.

fruit_type.sh × season

```
#!/bin/bash

echo "Enter fruit name:"
read fruit

case $fruit in
    apple|Apple|APPLE|mango|Mango|MANGO|banana|Banana|BANANA)
        echo "Common Fruits"
        ;;
    strawberry|Strawberry|STRAWBERRY|blueberry|Blueberry|BLUEBERRY)
        echo "Berries"
        ;;
    coconut|Coconut|COCONUT|walnut|Walnut|WALNUT)
        echo "Dry Fruits"
        ;;
    *)
        echo "Category Not Found"
        ;;
esac
```

```
student@Ubuntu1:~/DSY OS$ bash fruit_type.sh
Enter fruit name:
mango
Common Fruits
student@Ubuntu1:~/DSY OS$ bash fruit_type.sh
Enter fruit name:
strawberry
Berries
student@Ubuntu1:~/DSY OS$
```

7.

season.sh

```
#!/bin/bash

echo "Enter month name or number:"
read month

case $month in
    12|1|2|December|december|DECEMBER|January|january|JANUARY|February|february|FEBRUARY)
        echo "Winter"
        ;;
    3|4|5|March|march|MARCH|April|april|APRIL|May|may|MAY)
        echo "Spring"
        ;;
    6|7|8|June|june|JUNE|July|july|JULY|August|august|AUGUST)
        echo "Summer"
        ;;
    9|10|11|September|september|SEPTEMBER|October|october|OCTOBER|November|november|NOVEMBER)
        echo "Autumn"
        ;;
    *)
        echo "Invalid month."
        ;;
esac

```

```
student@Ubuntu1:~/DSY OS$ bash season.sh
Enter month name or number:
January
Winter
student@Ubuntu1:~/DSY OS$ bash season.sh
Enter month name or number:
10
Autumn
student@Ubuntu1:~/DSY OS$
```

Conclusion:

I learned how to use the case ... esac statement in shell scripting for multi-way decision making.
I learned to write and execute shell scripts using Linux commands and Bash.
I implemented programs to identify vowels, numbers, file types, characters, vehicles, fruits, and seasons.
Overall, this experiment helped me understand pattern matching and conditional execution in shell scripting.

Assessment Schemes:

Attendance	Discipline	Short oral	Correctness of Lab Report	Timely completion of Lab Report	Total marks (10)	Signature of Teacher with Date